

JOSE ERNESTO TOMANAN

Full-stack software developer

EDUCATION

University of the Philippines - Diliman 2022 – 2026
B.S. Computer Science, Cum Laude distinction (GWA 1.464)

Philippine Science High School - Central Luzon Campus 2015 – 2021
Junior & Senior High School (Physics Strand), Consistent Director's Lister

EXPERIENCE

Software Engineer Aug 2026 – present
Metro East Technology Resources, Inc.

- Using Go with Docker to build and maintain a microservices-based backend
- Focuses on optimizing for concurrency, resolving race conditions, & writing highly secure software

Backend Development Intern Jun 2025 – Jul 2025
Pointwest Innovations Corporation

- Tasked in a 10-man case study for developing an HR-based application for the company
- Built API services and controllers using Spring Boot; Developed testing suites to assure application functionality during development and refactoring

Tackle Co-Head Sep 2023 – Mar 2024
UP Association of Computer Science Majors (UP CURSOR)

- Co-headed a team of 20 in planning, marketing, and promoting a 4-day hybrid setup tech career fair.
- Led Branding & Creatives to design & market the event, and HR to internally disseminate the event

PROJECTS

SIPAT-MATH: AI Grading Automation for early-stage Mathematics Sep 2025 – May 2026
Capstone special project for Service Science and Software Engineering Laboratory (S3)

- Lead a 2-man capstone project in designing, developing, and evaluating a VLM-based tool that automates the assessment early-level Mathematics questions based on rubrics and answer keys
- Took charge of full-stack development using Svelte5 and FastAPI
- Won 3rd in Best Presentation (Design Category) in DCS KOLOKYAL 2026

PokéDex using Svelte Mar 2026
Pokémon species frontend interface

- Integrates the official PokéAPI with Svelte to create a responsive, mobile-friendly database interface with searching, sorting, & filtering functionality
- Uses SvelteKit 2 to handle API request handling & data management

JoseTomanan.GitHub.Io Jul 2024 – present
Personal portfolio webpage

- Online portfolio-resume presented as a single-page website created using Svelte and TailwindCSS
- Incorporates learned front-end tech into the site's interface, including use of libraries Iconify and Shadcn

Poverty and Parenting: A Data Science Study Feb 2024 – Jun 2024
Economic analysis of birth rates in the Philippines

- Used Python to show relationship between live births & poverty incidence across Philippine municipalities
- Utilized libraries: Pandas, Geopandas, Seaborn, using Jupyter Notebook

SKILLS

Programming Language: Python, Go, JavaScript, TypeScript, C

Libraries & Frameworks: Svelte5, SvelteKit, TailwindCSS, Spring Boot, Django

Technologies: Git, PostgreSQL, MySQL, Vite, Claude Code